

Exercise Sheet 9

Discrete Mathematics by Hongfei Fu, 2023.11.8

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1. Let R be a symmetric relation. Show that R^n is symmetric for all positive integers n .

Answer Area:

$\because R$ is a symmetric relation.

$\therefore R = R^{-1}$.

Assume that $R^n = (R^n)^{-1}$.

Then $R^{n+1} = R^n \circ R = (R^n)^{-1} \circ R^{-1} = (R \circ R^n)^{-1} = (R^n \circ R)^{-1} = (R^{n+1})^{-1}$.

$\therefore \forall n \in \mathbf{N}, R^n = (R^n)^{-1}$.

$\therefore R^n$ is symmetric.

2. Prove that for any relation R , we have $(R^*)^{-1} = (R^{-1})^*$.

Answer Area:

$$\bullet \forall x \forall y, x(R^*)^{-1}y, yR^*x.$$

$$\therefore \exists R^n \subseteq R^*, yR^n x.$$

$$\therefore x(R^n)^{-1}y.$$

$$\therefore (R^n)^{-1} = (R \circ R \circ \dots \circ R)^{-1} = R^{-1} \circ R^{-1} \circ \dots \circ R^{-1} = (R^{-1})^n.$$

$$\therefore x(R^{-1})^n y.$$

$$\therefore (R^{-1})^n \subseteq (R^{-1})^*.$$

$$\therefore x(R^{-1})^* y.$$

$$\bullet \forall x \forall y, x(R^{-1})^* y$$

$$\therefore \exists (R^{-1})^n \subseteq (R^{-1})^*, x(R^{-1})^n y.$$

$$\therefore (R^n)^{-1} = (R^{-1})^n.$$

$$\therefore x(R^n)^{-1} y.$$

$$\therefore yR^n x.$$

$$\therefore R^n \subseteq R^*.$$

$$\therefore yR^* x.$$

$$\therefore x(R^*)^{-1} y.$$

$$\therefore (R^*)^{-1} = (R^{-1})^*$$

3. Determine whether for any relation R, S , it holds that $(R \cup S)^* = R^* \cup S^*$? Please either prove or disprove your answer.

Answer Area:

No. An example is as follows:

Suppose that R and S are relations on $A = \{1, 2, 3, 4\}$.

$$R = \{(1, 2), (3, 4)\}, S = \{(1, 3), (2, 4)\}.$$

$$R^2 = R^3 = \dots = R^n =, S^2 = S^3 = \dots = S^n =$$

$$\text{Then } R^* = \{(1, 2), (3, 4)\}, S^* = \{(1, 3), (2, 4)\}, R^* \cup S^* = \{(1, 2), (3, 4), (1, 3), (2, 4)\}.$$

$$(R \cup S) = \{(1, 2), (3, 4), (1, 3), (2, 4)\}.$$

$$\text{Then } (R \cup S)^2 = \{(1, 4)\}, (R \cup S)^3 = \dots = (R \cup S)^n =$$

$$\text{Then } (R \cup S)^* = \{(1, 2), (3, 4), (1, 3), (2, 4), (1, 4)\}.$$

$$\therefore (R \cup S)^* \neq R^* \cup S^*$$

4. Given a relation R on a set A , determine whether there exists a smallest equivalence relation R' that includes R , i.e., satisfying the following conditions:

- $R \subseteq R'$;
- for all equivalence relations R'' , $R \subseteq R'' \Rightarrow R' \subseteq R''$.

Please either prove or disprove your answer.

Answer Area:

We can construct R' :

$$R' := \bigcap \{R'' \in 2^{A \times A} \mid R \subseteq R'' \text{ and } R'' \text{ is an equivalence relation.}\} =: \bigcap \mathcal{A}.$$

Note that $\mathcal{A}$ is non-empty since $A \times A$ is an equivalence relation and $R \in A \times A$. So $A \times A \in \mathcal{A}$. So, $\exists R' = \bigcap \mathcal{A}$.

Then we have:

- $\because \forall R'' \in \mathcal{A}, R \in R''$.
 $\therefore R \in \bigcap \mathcal{A} = R'$
- for all equivalence relations R'' , $R \subseteq R'' \Rightarrow R'' \in \mathcal{A} \Rightarrow R' = \bigcap \mathcal{A} \in R''$.
- R' is an equivalence relation since
 - ★ R' is reflexive:
 $\forall R'' \in \mathcal{A}, \forall a \in A, aR''a$.
 Then $\forall a \in A, aR'a$.
 - ★ R' is symmetric:

$$xR'y \Rightarrow \forall R'' \in \mathcal{A} (xR''y) \Rightarrow \forall R'' \in \mathcal{A} (yR''x) \Rightarrow yR'x;$$

- ★ R' is transitive:

$$(xR'y \cap yR'z) \Rightarrow \forall R'' \in \mathcal{A} (xR''y \cap yR''z) \Rightarrow \forall R'' \in \mathcal{A} (xR''z) \Rightarrow xR'z;$$

$\therefore$ There exists a smallest equivalence relation R' that includes R

5. Prove that for any relations R, S , we have that

$$R \circ S \subseteq S \Leftrightarrow (\bigcup_{n=1}^{\infty} R^n) \circ S \subseteq S.$$

Answer Area:

- If $R \circ S \subseteq S$. Assuming that $(x_0, y_0) \in (\bigcup_{n=1}^{\infty} R^n) \circ S$.
 $\therefore \exists z_0, x_0 S z_0 \cap z_0 R^* y_0$.
 $\therefore R \circ S \subseteq S$.
 $\therefore x R \circ S y \rightarrow x S y$.
 $\therefore z_0 R^* y_0$.
 $\therefore \exists R^n, z_0 R^n y_0$.
 $\therefore z_0 (R \circ R \circ \dots \circ R) y_0$
 $\therefore \exists y_1, y_2 \dots y_{n-1}, z_0 R y_1 R y_2 R \dots R y_{n-1} R y_0$.
 $\therefore x_0 S z_0 \cap z_0 R y_1$
 $\therefore x_0 R \circ S y_1$, then $x_0 S y_1$.
 Repeat the process above, $x_0 S z_0 \rightarrow x_0 S y_1 \rightarrow x_0 S y_2 \rightarrow \dots \rightarrow x_0 S y_0$.
 $\therefore (\bigcup_{n=1}^{\infty} R^n) \circ S \subseteq S$.

- If $(\bigcup_{n=1}^{\infty} R^n) \circ S \subseteq S$:
 $\because R \subseteq (\bigcup_{n=1}^{\infty} R^n)$.
 $\therefore R \circ S \subseteq (\bigcup_{n=1}^{\infty} R^n) \circ S \subseteq S$.

$$\therefore R \circ S \subseteq S \Leftrightarrow (\bigcup_{n=1}^{\infty} R^n) \circ S \subseteq S .$$